PRIOR-ART REVIEW | RESEARCH DATE: 25 OCTOBER 2026

Offline Multimodal Biometrics: Has It Already Been Built?

Face + voice verification, a random spoken challenge, protected on-device templates, and proven scalability for 1:1 verification - checked against 14 publicly documented systems and 29 sources.

Verdict: Only partial prior art found - no public exact match.
No publicly documented system was found that satisfies all five target criteria at the same time. Offline face+voice exists. Random spoken challenges exist. Protected on-device templates exist. But no system combines all of them - with measured scalability for the complete system.

The three closest systems each miss something different

- **Sensory TrulySecure** - fully on-device face+voice with encrypted / Trusted Execution Environment (TEE) templates. Misses: its voice modes are fixed or user-defined passphrases, not a fresh random challenge, and it publishes no user-scaling evaluation.
- **MobiDev Jetson entrance prototype** - face identification plus 1:1 voice verification of a randomly generated phrase, all on one edge device. Misses: face and voice vectors sit in a plain SQLite database with no encryption described, and there is no user-scaling evaluation.
- **Revay et al. (2026)** - face + speaker verification with a random spoken challenge, processed on-premise. Misses: it is a client-server web application, template encryption is not described, and the authors state it was not subjected to large-scale evaluation.

> **What this means for the prototype:** the components are proven, but the integrated combination - and its combined evaluation - is the open gap. That gap is the contribution, provided the claims stay limited to what is publicly documented and later measured.

Prepared for Kanika - multimodal biometrics research (offline and scalable direction)
Sources: academic papers, official product and SDK pages, reputable trade press, and vendor engineering write-ups. Each source is flagged *verified live* (opened during research) or *index* (search-index text only). Full source list on pages 13-14.

The five criteria the target system must meet together

A system counts as an exact match only if it meets all five at once. Meeting three or four is close - and several systems do - but it is not the same system.

1. **Face and voice together.** Face verification and speaker/voice verification used together, preferably from one capture or session - not two separate products sold side by side.
2. **A random spoken challenge.** A fresh, randomly generated phrase or number sequence for freshness and liveness, where both the spoken content *and* the speaker's voice are checked. A fixed passphrase does not count.
3. **Fully offline verification.** Enrolment templates and all matching stay local or on-device, with no network required during verification. On-premise server processing is local to a site, but it is not on-device.
4. **Protected local templates.** Templates are encrypted or otherwise protected where they are stored locally - not plain files or an unencrypted database.
5. **Scalability evidence for 1:1 verification.** An explicit design or evaluation as user numbers grow: constant-time per-user matching demonstrated, or enrolment, storage, latency, memory or load benchmarks as enrolment count varies.

> **How criterion 5 was judged:** a single-verification latency figure such as “under 1 second” or “2,719 ms” is **not** scalability evidence. Scalability evidence varies the enrolled population and reports latency, storage, memory or throughput against it, runs a load test, or explicitly designs 1:1 claimed-identity retrieval so matching cost does not grow with the population. Almost no source does this for an offline face+voice system.

How to read the ratings in this document

Yes = documented in the source. **Partial** = partial, qualified, or asserted without numbers. **No** = not documented, or the source describes the opposite. Ratings judge *public documentation* only - a closed SDK may do more than its public pages describe (see “Could not verify”, page 12).

Closest match 1: Sensory TrulySecure

Commercial SDK | Meets criteria 1, 3, 4 | Misses 2 and 5

TrulySecure is the strongest commercial match. It is a real, FIDO-certified SDK with simultaneous face and voice enrolment, fully on-device matching, and on-device encryption or TEE storage.

What is documented

- **Face + voice, one capture:** enrolment is “capturing voice and face simultaneously in a few seconds. Authentication is on-device and almost instantaneous”, and the speaker SDK can be “configured to require both face and voice confirmation for greater user security (TrulySecure mode)”.
- **Fully offline:** the FIDO Alliance showcase describes it as “an on-device biometric authenticator that does not require a cloud connection”. Sensory's current solution page, verified live on 25 October 2026, states: “Secure face and voice verification that stays entirely on the device”, “Works with or without network”, and “No dependency on remote servers”.
- **Protected templates:** “All processing and encryption is done on-device, so personal data remains secure.” Sensory's CEO, quoted in EDN: “We can encrypt the biometric representation and we can store it on a device on the Trusted Execution Environment (TEE).”

Why it is not an exact match

- **The spoken challenge is not random.** The documented text-dependent voice modes are only a fixed passphrase (predefined) and a user-defined passphrase, plus text-independent verification (any speech). Its “voice liveness” is described as an interactive challenge sequence, but not as a fresh random phrase whose words are transcribed and checked. Its later face-liveness challenge is movement-based - following prompts about what movement to make.
- **No scalability evaluation.** Public figures are a 10-second enrolment and under 1 second to authenticate, plus accuracy testing on data from “thousands of actual users”. That is an accuracy cohort, not latency, storage or memory measured as enrolment grows, and there is no load test.
- **Configuration caveat:** TrulySecure 4.0 also offers a federated/cloud mode in which an on-device enrolment template “can be handed off to the authentication server” - so not every configuration is purely local.

> **Bottom line:** TrulySecure proves offline face+voice with protected templates is a shipped product. It does not prove the random-challenge, scalable version of that system.

Sources: sensory.com/solution/pc-mobile-tablet (verified live); fidoalliance.org/showcase/trulysecure; sensory.com TrulySecure 2.0 releases; edn.com blending-biometrics article. Full links in Sources [1], [3]-[9].

Closest matches 2 and 3: a random challenge, without protection or scale

MobiDev Jetson entrance system - vendor prototype

Meets 1, 2, 3 (single edge device) **Misses 4 and 5**

- **Design:** face identification first (RetinaFace + SE-ResNet-50 against an employee photo), then voice verification (ECAPA-TDNN via SpeechBrain, trained on VoxCeleb2). Voice is verified 1:1 against the face-identified person, in the same entrance session.
- **Random spoken challenge:** “when each person verifies themselves, they pronounce a phrase given by the system. The phrase consists of randomly selected words of a chosen set.” Speech-to-text (QuartzNet, Levenshtein distance) checks the content while the speaker model checks the voice. Recommended utterance: at least 3 seconds, about 8 words.
- **Local:** all four models run on a single NVIDIA Jetson Xavier that is both sensor and door controller, and the device “can operate even in case of connection shut down”.
- **Why it falls short:** “This data currently includes precomputed face and voice vectors for each user... we've chosen SQLite DB” - no encryption or template protection is described anywhere. Its only scaling statement is adding more devices per location. The face step is also 1:N identification, whose cost ordinarily grows with gallery size, and the article does not address that.

Revay, Kanuk and Bencik (2026) - academic prototype, Science & Military

Meets 1, 2, local on-premise processing **Misses strict on-device, 4, 5**

- **Design:** “face recognition with liveness detection, role-based access control, and a second authentication factor based on speaker verification with a spoken challenge-response mechanism... all biometric processing is performed locally within on-premise infrastructure.”
- **Random challenge, both checks:** “the system generates a random spoken challenge... Two parallel checks are performed: verification of the spoken challenge content and speaker verification based on voice embeddings.” The challenge is a dynamically generated numerical challenge spoken aloud.
- **Why it falls short:** it is a client-server web application - “the server component constitutes the computational core” and the browser client only handles interaction and capture. Its storage section describes separating records from templates and not retaining raw data, but states no encryption mechanism. On scale, the authors are explicit: the prototype “was not subjected to large-scale statistical evaluation”, and a larger population is future work.

Closest matches 4 and 5: strong ideas, missing deployment evidence

Jha, Jain and Srivastava (2025) - challenge-response paper

Paper, Scalable Computing: Practice and Experience **Meets 1, 2 as an algorithm study**

- “The challenge-response approach requires the user to utter a random word. Next, the system detects the user's facial features (eye and mouth motion) and recognized speech text to confirm whether the authentication request originates from a legitimate user or an imposter.”
- Feature-level fusion with a CNN-BiLSTM on a self-built database of **55 users**, achieving 96.81% accuracy, 98.20% precision, and an Equal Error Rate (EER) of 3.37%.
- **Missing:** no on-device or offline deployment architecture, no encrypted local templates, and no scaling evaluation. Despite the journal's name, the paper contains no scalability study - 55 users is an accuracy test set, not a scalability benchmark.

de la Torre et al. (2026) - ESP32 prototype, Sensors

Prototype paper **Meets 1, 3** **Misses 2, 4 (explicitly), 5**

- Fully standalone: hierarchical person detection, then voice, then face on an ESP32, with a combined False Acceptance Rate (FAR) of 0.12% and a “privacy-by-design architecture that completely eliminates the need for raw data transmission to cloud servers”. Energy use is 0.15 J per inference cycle - about 39 hours of continuous inference on 600 mAh.
- **Not a random challenge, and not template 1:1:** the voice module is a classifier whose authorised class is 100 captures of one target user speaking two fixed passphrases (“Hi, I am [Name]” and “Open”). The authors note this “inherently limits the generalization capabilities of the models”.
- **Fails criterion 4 explicitly:** “In our current proof-of-concept, the stored neural network weights and authorized user biometric templates (embeddings) reside in unencrypted Flash memory.” Protection is proposed only for a future commercial deployment.
- **Benchmarks are per-inference, not user-scaling:** end-to-end verification in 2,719 ms with a maximum RAM footprint of 377.3 KiB (face module, INT8: 1,102 ms). Enrolment count is never varied - the population is one authorised user.

> **Pattern across all five closest systems:** each one proves a different subset. None was found that adds the missing pieces - random challenge *and* protected local templates *and* offline matching *and* a scaling evaluation - in a single documented system.

Most multimodal systems run on a server or in the cloud

Representative face+voice systems that are server- or cloud-based, not fully offline, or lack a documented random spoken challenge:

System	What it does	Why it is not the target
Daon IdentityX	Hosted platform: face, voice, fingerprint, behavioural, palm; FIDO authenticators.	Server components are “responsible for the secure storage and matching”; the client only captures. No random spoken challenge documented. Its scaling claim is server throughput (page 9).
Aware Knomi	Face+voice, 1:1 and 1:N, passive liveness; device, server and browser configurations.	Knomi S “creates, matches, and stores biometric templates on server platforms”; the Face Authentication Service is a server-based API. No random challenge or local encrypted-template evidence on the page.
BioID Web Service	Cloud “Biometrics as a Service” via APIs; Verify API performs a 1:1 comparison of uploaded samples with a stored template.	Cloud API requiring uploads. Historical multimodal description is explicitly a “cloud-based online service... with dynamic scaling”. Current positioning is face-centric. Not offline; templates not local.
ImageWare Authenticate	Enterprise multimodal platform; aggregator description includes voice where “users speak a randomized phrase”, with two or more modalities enforceable.	Has criteria 1+2 on that description, but no evidence of offline on-device matching, local protected templates, or a scaling evaluation was found. Aggregator-sourced only.
Neurotechnology MegaMatcher ID	SDK + web service for face, voice and contactless fingerprint; encrypted enrolment data; data “can be kept on the end-user device”.	Components *can* be assembled on-device, but no random spoken challenge is documented - voice liveness uses a phrase selected from an enrolled set, and text-dependent verification uses the same phrase at enrolment and verification. A sibling voice product also requires server accounting contact. No user-scaling evaluation for the on-device configuration.
USAA mobile banking (historical)	Face (look + blink) and voice (read a short phrase) plus PIN in a banking app.	Server-backed (built on Daon IdentityX). The voice phrase is a read phrase, not documented as a fresh random challenge verified locally. Not an offline system.

Sources [14]-[24], [29] in the source list. Vultus, which pairs a genuinely random phrase with encrypted templates but runs matching on its server, is covered in detail on the next page.

Random phrase + encryption exists today - as a cloud service

Vultus (CYBERVIS)

Vultus is instructive because it gets the challenge right and still misses the target on architecture. “After the face check, the user repeats a randomized phrase - Vultus matches both the words and their voiceprint, so a replayed recording fails.” Only an encrypted template is stored, isolated per application (AES at rest), and the voiceprint is likewise encrypted.

But “face matching and liveness run on the frames in RAM on the Vultus server”, hosting is on CYBERVIS UK infrastructure by default (or inside the customer's own network), and every sign-in produces a “short-lived signed token that your backend verifies”. Plan tiers of 50 / 1,000 / 10,000 enrolled faces are commercial caps, not benchmarks. Vultus meets criteria 1, 2 and 4 (server-side) and fails 3 and 5.

On-device systems that are unimodal, or only components

- **Apple Face ID (face only).** Face data is “encrypted and protected by the Secure Enclave... doesn't leave your device, is never available to iOS or iPadOS, apps, or Apple, and is never backed up to iCloud”. There is no voice modality and no spoken challenge, and an app receives only the operating system's yes/no - so it cannot be used to build the target system.
- **FacePing (face-only SDK, verified live).** “Face verification that runs on the phone, works with no signal and has liveness built in”; under 0.3 s per check; “checks need no network at all” after an initial sync; “Face templates are sealed with AES-256-GCM, with the key in the Keychain or Keystore”; by default it “checks each person against their own face (1:1)”. This is the clearest public statement of the 1:1 claimed-identity design - but it is face-only, templates are also held on the vendor's servers and synced, and its 5,000 / 30,000-people tiers are billing units, not latency benchmarks.
- **Component SDKs and models (not systems).** Neurotechnology VeriSpeak / VeriLook, Aware Nexa Voice / Nexa Face, SpeechBrain ECAPA-TDNN and InsightFace each provide one modality or one algorithm. None by itself documents enrolment-to-verification for face+voice with a random challenge, offline operation and scaling. Treating a component catalogue as a built multimodal system would be a category error.

Products that do face + voice offline - and their limits

Three commercial answers were found. No open-source equivalent was found in the searched sources; open-source results were single-modality libraries and models only.

1. Sensory TrulySecure / AppLock by Sensory

The clearest commercial answer (covered on page 3). In AppLock, users choose “three pre-determined voice passphrases or... their own user-defined passphrase”, repeated three times at enrolment, and TrulySecure mode “requires verification of both face and voice biometrics” on-device. Limits: fixed or user-defined phrase, no random challenge, no scaling evaluation.

2. OneFaceIn (OFI) Biometric Password Manager

A consumer app: face data points “locally stored on user devices” with liveness against still images; voice enrolment is “repeat a phrase three times consecutively”; the multimodal prompt centres the face and captures audio together. App-listing copy adds that it “does not require a web based connection as everything is synced to your device and locally encrypted and decrypted”. Limits: marketing and app-store copy only, no independent evaluation, a fixed phrase, no scalability evidence, and dated (2016-era) material.

3. Neurotechnology MegaMatcher ID

An SDK, not a finished product. It *can* be configured for on-device face+voice (page 6), so an integrator could build such an app - but Neurotechnology does not document a finished offline face+voice product with a random spoken challenge or a scaling evaluation.

No other commercial product in the searched sources was documented as performing both face and voice verification entirely offline on the device. Other face+voice products found (Daon, Aware, BioID, Vultus, ImageWare, Wink) are server or cloud architectures in their documented configurations.

Patents are designs on paper, not built systems

- **EP3156978A1, “secure speaker verification”:** “generating a random, textual prompt for a user to speak... recording ultrasound signals reflected from the user's face... detecting liveness based on correspondence of audio with ultrasound.” The face here is an ultrasound liveness signal, not face-template verification. No offline architecture or scaling is evidenced.
- **US20180060552A1, voice-based liveness verification:** challenge prompts are “liveness passphrases... selected randomly from a larger set of liveness passphrases that have been captured from an enrolled user”, combinable with a fingerprint scan or face capture. A patent application, not a documented product or evaluation.
- Randomized phrase prompting itself is old - Higgins et al., “Speaker Verification Using Randomized Phrase Prompting”, is cited in later patents. What was not found is its combination with criteria 1, 3, 4 and 5 in one built, evaluated, offline system.

Scalability: the strongest evidence belongs to server systems

1:1 verification vs 1:N identification. In 1:1 verification the user claims an identity, the system retrieves that one enrolled template, and performs one comparison per modality - cost is in principle independent of the total enrolled population (storage still grows linearly, one template per user). In 1:N identification the probe is compared against the whole gallery, so cost grows with population unless indexing, approximate search or server accelerators are used. A scalability claim is only meaningful once it says which mode, what was varied, and what was measured.

Source	Mode / architecture	What was measured or stated	Assessment
Daon IdentityX	Server platform	“Scaled to populations of hundreds of millions. USAA has reported... 750,000 authentications in a day”	Real population + throughput - for a server system. Says nothing about offline on-device 1:1 cost.
Neurotechnology MegaMatcher ABIS	Server 1:N	“Up to 1.2 billion comparisons per second on each node”; national-scale registries	Genuine 1:N throughput benchmark - server nodes, the opposite deployment from the target.
Sensory TrulySecure	On-device 1:1	Under 1 s authenticate; 10 s enrolment; accuracy on data from thousands of users	Single-transaction latency + accuracy cohort. Enrolment never varied. **Not scalability evidence.**
FacePing	On-device 1:1 by design	Under 0.3 s per check; explicit 1:1 design; pricing to 30,000 stored people	Design statement + per-check latency. No latency/storage curve vs enrolled count. Face only.
ESP32 paper	On-sensor, 1 user	2,719 ms, 377.3 KiB RAM, 0.15 J per inference	Careful benchmarks at a fixed population of one. Feasibility, not scaling.
Revay et al. / Jha et al. / MobiDev	On-premise / algorithm / edge	No large-scale evaluation (Revay); 55-user accuracy (Jha); “scaled to locations” by adding devices (MobiDev)	Explicit absence, an accuracy sample, and a location assertion. None varies enrolled users.
BioID / Vultus / Aware	Cloud / server	“Dynamic scaling”; plan caps to 10,000 faces; “1:1 and 1:N” feature listing	Marketing assertions and commercial caps, with no published benchmark in the sources read.

Bottom line on criterion 5: no source found benchmarks an offline face+voice system as enrolled users grow - no latency-vs-N, storage-vs-N, memory-vs-N, concurrent-user load test, or measured constant-time 1:1 result for the combined system.

Comparison matrix - systems 1 to 7 of 14

Yes = documented. Partial = partial, qualified, or assertion only. No = not documented / contrary evidence. “Random challenge” means a fresh random spoken phrase whose content and speaker are both checked.

System	Face + Voice	Random challenge	Fully offline	Protected local templates	Scalability (1:1)	Key limitation
Sensory TrulySecure (SDK)	Yes	No	Yes	Yes	No	Fixed / user-defined passphrase; under 1 s single auth only; 4.0 also offers a server mode.
MobiDev Jetson (prototype)	Yes	Yes	Yes	No	No	Vectors in plain SQLite; scaling = adding devices; face step is 1:N; vendor blog.
Revay et al. 2026 (paper)	Yes	Yes	Partial	No	No	Client-server on-premise, not on-device; encryption not stated; no FAR/FRR or scaling statistics.
Jha et al. 2025 (paper)	Yes	Yes	No	No	No	Algorithm study on 55 users; no deployed offline system.
de la Torre et al. 2026 (ESP32 paper)	Yes	No	Yes	No	Partial	Two fixed passphrases; templates explicitly unencrypted in Flash; population = 1 user.
MegaMatcher ID (SDK)	Yes	No	Partial	Yes	No	Phrase-set challenge, same-phrase verification; component system, not a finished offline product.
OneFaceIn (consumer app)	Yes	No	Yes	Yes	No	Fixed phrase repeated 3x; marketing copy only; dated; no evaluation.

Comparison matrix - systems 8 to 14 of 14

System	Face + Voice	Random challenge	Fully offline	Protected local templates	Scalability (1:1)	Key limitation
Vultus (web service)	Yes	Yes	No	Partial	No	Matching on Vultus server; encrypted templates are server-side; voice is an optional add-on.
ImageWare Authenticate (platform)	Yes	Yes	No	No	No	Aggregator-sourced only; enterprise/server positioning; primary docs not captured.
Daon IdentityX (server platform)	Yes	No	No	Partial	Partial	Server stores and matches; its scaling (hundreds of millions; 750k/day) is server throughput - the wrong architecture for this target.
Aware Knomi (framework)	Yes	No	No	No	No	Documented configurations are server-centric; 1:1 and 1:N listed with no numbers.
BioID BWS (cloud service)	Yes	No	No	No	No	Biometrics-as-a-Service cloud model; samples uploaded to BWS storage.
Apple Face ID (platform)	No	No	Yes	Yes	No	Face only; device-owner templates; cannot provide custom speaker verification to apps.
FacePing (SDK)	No	No	Yes	Yes	Partial	Face only; explicit 1:1-by-design statement but no latency-vs-N curve; enrolment/sync needs network.

Reading the matrix: no row is “Yes” across all five criteria. The two rows with a random challenge and offline/local processing (MobiDev, Revay) both lack protected templates and scaling evidence. The rows with protected local templates and offline operation (Sensory, OneFaceIn, Face ID, FacePing) lack either a random challenge, the voice modality, or both.

What could not be verified - read the verdict with these limits

- **Sensory's “interactive challenge sequence” for voice liveness:** public material does not specify whether the sequence is random, spoken, or how it is checked, and no SDK documentation with a random-phrase mode was publicly accessible. It is scored “No” for criterion 2 - not asserted to be impossible in the closed SDK.
- **ImageWare Authenticate:** only aggregator descriptions were captured; its primary documentation was not opened. Its deployment model is “no offline evidence found”, not a proven negative.
- **Revay et al. template encryption:** the paper states separation of records and templates and non-retention of raw data, with no encryption statement. The claim here is only that protection is *not evidenced* in the public paper.
- **Closed SDK internals generally:** TrulySecure, Knomi, MegaMatcher ID and similar SDKs are proprietary. Absence of a feature in public marketing or documentation is not proof the SDK cannot be configured for it. The verdict is strictly about what is publicly documented.
- **Open-source exact match:** searches surfaced only single-modality libraries and models. No open-source repository documenting the full five-criterion system was found; this is a search result, not an exhaustive audit of all repositories.

What this means for the offline, scalable prototype

- The defensible claim is **scalable 1:1 multimodal verification** - not 1:N identification from a growing gallery, which is a different and harder claim.
- The contribution is the **combination plus its measurement**: face + voice + a fresh random spoken challenge, templates protected on the device, zero network during verification, and authentication time, storage and memory actually measured as enrolment grows (for example 1,000 to 100,000 enrolled users).
- The safe wording is: *“No publicly documented exact match was found in the searched sources.”* Do not shorten that to “nobody has ever built this” - closed commercial SDKs could contain undocumented capabilities.
- Banking remains only an example application of this system. The prototype aim is the offline and scalable multimodal system itself.

Sources (1 of 2)

All read 25 October 2026. “Verified live” = page opened during research. “Index” = search-index text only.

1. Sensory, PC/Mobile/Tablet solution page - face+voice on-device, fully offline - https://sensory.com/solution/pc-mobile-tablet/ - **verified live**
2. FacePing - offline face-only SDK, AES-256-GCM, 1:1 by default - https://faceping.ai/ - **verified live**
3. FIDO Alliance, TrulySecure showcase - https://fidoalliance.org/showcase/trulysecure/ - index
4. Sensory, TrulySecure Speaker Verification 2.0 - https://sensory.com/news/sensory-adds-suite-of-new-biometric-voice-features-to-trulysecure-speaker-verification/ - index
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17. BioID homepage - https://www.bioid.com/ - index
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19. Vultus (CYBERVIS) - https://vultus.cybervis.co.uk/ - index (full text)
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Also cited in the text: Higgins et al., “Speaker Verification Using Randomized Phrase Prompting” (cited in later patents, for the history of randomized phrase prompting), and US20180060552A1, voice-based liveness verification (Google Patents).

This document is the downloadable version of the prior-art comparison delivered on 25 October 2026. It preserves the research report's verdict and evidence. Where public documentation was silent, this document says so rather than treating silence as proof that a capability does not exist.